

# Homework Set 3:-

**11A.1. Temperature in a friction bearing.** Calculate the maximum temperature in the friction bearing of Problem 3A.1, assuming the thermal conductivity of the lubricant to be  $4.0 \times 10^{-4} \text{ cal/s} \cdot \text{cm} \cdot ^\circ\text{C}$ , the metal temperature  $200^\circ\text{C}$ , and the rate of rotation 4000 rpm.  
**Answer:** About  $225^\circ\text{C}$

**3A.1 Torque required to turn a friction bearing** (Fig. 3A.1). Calculate the required torque in  $\text{lb} \cdot \text{ft}$  and power consumption in horsepower to turn the shaft in the friction bearing shown in the figure. The length of the bearing surface on the shaft is 2 in, and the shaft is rotating at 200 rpm. The viscosity of the lubricant is 200 cp, and its density is  $50 \text{ lb}_m/\text{ft}^3$ . Neglect the effect of eccentricity.  
**Answers:**  $0.32 \text{ lb} \cdot \text{ft}$ ;  $0.012 \text{ hp}$  =  $0.009 \text{ kW}$

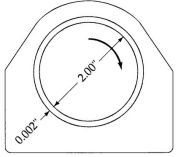


Fig. 3A.1. Friction bearing.

changing along the  $y$  direction:  
 $T = T(y)$

## §B.9 THE EQUATION OF ENERGY FOR PURE NEWTONIAN FLUIDS WITH CONSTANT $\rho$ AND $k$

Cartesian coordinates  $(x, y, z)$ :

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v \quad (\text{B.9-1})^b$$

Cylindrical coordinates  $(r, \theta, z)$ :

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v \quad (\text{B.9-2})^b$$

Spherical coordinates  $(r, \theta, \phi)$ :

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial T}{\partial \phi} \right) = k \left[ \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} \right] + \mu \Phi_v \quad (\text{B.9-3})^b$$

<sup>a</sup> This form of the energy equation is also valid under the less stringent assumptions  $k = \text{constant}$  and  $(\partial \ln \rho / \partial \ln T)_p Dp/Dt = 0$ . The assumption  $\rho = \text{constant}$  is given in the table heading because it is the assumption more often made.

<sup>b</sup> The function  $\Phi_v$  is given in §B.7. The term  $\mu \Phi_v$  is usually negligible, except in systems with large velocity gradients.

## §B.7 THE DISSIPATION FUNCTION $\Phi_v$ FOR NEWTONIAN FLUIDS (SEE EQ. 3.3-3)

Cartesian coordinates  $(x, y, z)$ :

$$\Phi_v = 2 \left[ \left( \frac{\partial v_x}{\partial x} \right)^2 + \left( \frac{\partial v_y}{\partial y} \right)^2 + \left( \frac{\partial v_z}{\partial z} \right)^2 \right] + \left[ \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right]^2 + \left[ \frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right]^2 + \left[ \frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right]^2 - \frac{2}{3} \left[ \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right]^2 \quad (\text{B.7-1})$$

Cylindrical coordinates  $(r, \theta, z)$ :

$$\Phi_v = 2 \left[ \left( \frac{\partial v_r}{\partial r} \right)^2 + \left( \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\theta}{r} \right)^2 + \left( \frac{\partial v_z}{\partial z} \right)^2 \right] + \left[ r \frac{\partial}{\partial r} \left( \frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]^2 + \left[ \frac{1}{r} \frac{\partial v_r}{\partial \theta} + \frac{\partial v_\theta}{\partial r} \right]^2 + \left[ \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right]^2 - \frac{2}{3} \left[ \frac{1}{r} \frac{\partial}{\partial r} (rv_r) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \right]^2 \quad (\text{B.7-2})$$

Spherical coordinates  $(r, \theta, \phi)$ :

$$\Phi_v = 2 \left[ \left( \frac{\partial v_r}{\partial r} \right)^2 + \left( \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\theta}{r} \right)^2 + \left( \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_\phi}{r} \cot \theta \right)^2 \right] + \left[ r \frac{\partial}{\partial r} \left( \frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]^2 + \left[ \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left( \frac{v_\phi}{\sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} \right]^2 + \left[ \frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} + r \frac{\partial}{\partial r} \left( \frac{v_\phi}{r} \right) \right]^2 - \frac{2}{3} \left[ \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} \right]^2 \quad (\text{B.7-3})$$

$$0 = k \frac{d^2 T}{dy^2} + \mu \left( \frac{dv_x}{dy} \right)^2$$

$$v_x \left( \frac{y}{h} \right)$$

$$\frac{dv_x}{dy} = \frac{v}{h}$$

$$k \frac{d^2 T}{dy^2} + \mu \left( \frac{v}{h} \right)^2 = 0$$

integrate twice:

$$\frac{dT}{dy} = -\frac{\mu U^2}{k h^2} y + C_1$$

$$T = -\frac{\mu U^2}{2 k h^2} y^2 + C_1 y + C_2$$

B.C. 1 :  $T(0) = 200$

B.C. 2 :  $T(h) = 200$

$$C_1 = \mu U^2 / 2 k h$$

$$C_2 = 200$$

$$T = -\frac{\mu U^2}{k h^2} y + \frac{\mu U^2}{2 k h} = 0$$

$$y = \frac{h}{2} \quad y = h/2$$

$$T_{\max} = 200 + \mu U^2 / 8 k$$

$$\omega = 2\pi \cdot 4000 / 60 = 419 \text{ rad/s}$$

$$U = \omega \times R = 225^\circ\text{C}$$

### 11B.2. Viscous heating in laminar tube flow (asymptotic solutions).

(a) Show that for fully developed laminar Newtonian flow in a circular tube of radius  $R$ , the energy equation becomes

$$\rho \hat{C}_p v_{z,\max} \left[ 1 - \left( \frac{r}{R} \right)^2 \right] \frac{\partial T}{\partial z} = k \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + \frac{4\mu v_{z,\max}^2}{R^2} \left( \frac{r}{R} \right)^2 \quad (11B.2-1)$$

if the viscous dissipation terms are not neglected. Here  $v_{z,\max}$  is the maximum velocity in the tube. What restrictions have to be placed on any solutions of Eq. 11B.2-1?

(b) For the *isothermal wall* problem ( $T = T_0$  at  $r = R$  for  $z > 0$  and at  $z = 0$  for all  $r$ ), find the asymptotic expression for  $T(r)$  at large  $z$ . Do this by recognizing that  $\partial T / \partial z$  will be zero at large  $z$ . Solve Eq. 11B.2-1 and obtain

$$T - T_0 = \frac{\mu v_{z,\max}^2}{4k} \left[ 1 - \left( \frac{r}{R} \right)^4 \right] \quad (11B.2-2)$$

(c) For the *adiabatic wall* problem ( $q_r = 0$  at  $r = R$  for all  $z$ ) an asymptotic expression for large  $z$  may be found as follows: Omit the heat conduction term in Eq. 11B.2-1, and then average the remaining terms over the tube cross section, by multiplying by  $r dr$  and then integrating from  $r = 0$  to  $r = R$ . Then integrate the resulting equation over  $z$  to get

$$\langle T \rangle - T_1 = (4\mu v_{z,\max} / \rho \hat{C}_p R^2) z \quad (11B.2-3)$$

in which  $T_1$  is the inlet temperature at  $z = 0$ . Postulate now that an asymptotic temperature profile at large  $z$  is of the form

$$T - T_1 = (4\mu v_{z,\max} / \rho \hat{C}_p R^2) z + f(r) \quad (11B.2-4)$$

Substitute this into Eq. 11B.2-1 and integrate the resulting equation for  $f(r)$  to obtain

$$T - T_1 = \frac{4\mu v_{z,\max}}{\rho \hat{C}_p R^2} z + \frac{\mu v_{z,\max}^2}{k} \left[ \left( \frac{r}{R} \right)^2 - \frac{1}{2} \left( \frac{r}{R} \right)^4 \right] \quad (11B.2-5)$$

Keep in mind that the solutions in Eqs. 11B.2-2 and 5 are valid solutions only for large  $z$ . The complete solution for small  $z$  is discussed in Problem 11D.2.

Assuming: S.S

Constant  $\nu$  and  $\mu$

Laminar flow:

$$v_z(r) = v_{z,\max} \left[ 1 - \left( \frac{r}{R} \right)^2 \right], \quad v_{z,\max} = \frac{(P_0 - P_L) R^2}{4\mu L}$$

$$T = T(r, z)$$

### a) Eqn for cylindrical coordinates.

#### §B.9 THE EQUATION OF ENERGY FOR PURE NEWTONIAN FLUIDS WITH CONSTANT<sup>a</sup> $\rho$ AND $k$

$$[\rho \hat{C}_p DT/Dt = k \nabla^2 T + \mu \Phi_v]$$

Cartesian coordinates ( $x, y, z$ ):

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v \quad (B.9-1)^b$$

Cylindrical coordinates ( $r, \theta, z$ ):

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + v_\theta \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v \quad (B.9-2)^b$$

Spherical coordinates ( $r, \theta, \phi$ ):

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial T}{\partial \phi} \right) = k \left[ \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} \right] + \mu \Phi_v \quad (B.9-3)^b$$

<sup>a</sup> This form of the energy equation is also valid under the less stringent assumptions  $k = \text{constant}$  and  $(\partial \ln \rho / \partial \ln T) D\rho/Dt = 0$ . The assumption  $\rho = \text{constant}$  is given in the table heading because it is the assumption more often made.

<sup>b</sup> The function  $\Phi_v$  is given in §B.7. The term  $\mu \Phi_v$  is usually negligible, except in systems with large velocity gradients.

#### §B.7 THE DISSIPATION FUNCTION $\Phi_v$ FOR NEWTONIAN FLUIDS (SEE EQ. 3.3-3)

Cartesian coordinates ( $x, y, z$ ):

$$\Phi_v = 2 \left[ \left( \frac{\partial v_x}{\partial x} \right)^2 + \left( \frac{\partial v_y}{\partial y} \right)^2 + \left( \frac{\partial v_z}{\partial z} \right)^2 + \left( \frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right)^2 + \left( \frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right)^2 + \left( \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right)^2 \right] \quad (B.7-1)$$

Cylindrical coordinates ( $r, \theta, z$ ):

$$\Phi_v = 2 \left[ \left( \frac{\partial v_r}{\partial r} \right)^2 + \left( \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\theta}{r} \right)^2 + \left( \frac{\partial v_z}{\partial z} \right)^2 + \left[ r \frac{\partial}{\partial r} \left( \frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]^2 + \left[ \frac{1}{r} \frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \right]^2 + \left[ \frac{1}{r} \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right]^2 \right] \quad (B.7-2)$$

Spherical coordinates ( $r, \theta, \phi$ ):

$$\Phi_v = 2 \left[ \left( \frac{\partial v_r}{\partial r} \right)^2 + \left( \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_\theta}{r} \right)^2 + \left( \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_\phi}{r} \cot \theta \right)^2 + \left[ r \frac{\partial}{\partial r} \left( \frac{v_\theta}{r} \right) + \frac{1}{r} \frac{\partial v_r}{\partial \theta} \right]^2 + \left[ \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left( \frac{v_\phi}{\sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} \right]^2 + \left[ \frac{1}{r \sin \theta} \frac{\partial v_z}{\partial \phi} + r \frac{\partial}{\partial r} \left( \frac{v_\phi}{r} \right) \right]^2 + \left[ \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_\theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (v_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} \right]^2 \right] \quad (B.7-3)$$

$$\rho \hat{C}_p v_z \frac{\partial T}{\partial z} = k \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + \mu \left( \frac{\partial v_z}{\partial r} \right)^2$$

$$\frac{\partial v_z}{\partial r} = - \frac{2 v_{\max}}{R^2} r = -A r$$

$$\rho \hat{C}_p v_{z,\max} \left[ 1 - \left( \frac{r}{R} \right)^2 \right] \frac{\partial T}{\partial z} = k \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + 4\mu \frac{v_{z,\max}^2}{R^2} \left( \frac{r}{R} \right)^2$$

b) for a large  $z$ ,  $\Delta z \rightarrow \infty$ ,  $\frac{\partial T}{\partial z} = 0$

$$\frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) = -4 \frac{\mu v_{z,\max}^2}{k R^2} \left( \frac{r}{R} \right)^2$$

$$\text{B.C.1: } r=0, T=\infty$$

$$\text{B.C.2: } r=R, T=T_0$$

integrating the eqn

$$\frac{\partial T}{\partial r} = - \frac{\mu v_{z,\max}^2}{k R^2} r^3 + \frac{C_1}{r}$$

integrating again:-

$$T = - \frac{\mu v_{z,\max}^2}{4k R^2} r^4 + \frac{C_1}{r^2} + C_2$$

$$\text{B.C.1: } 0 = 0 + \frac{C_1}{r} \rightarrow C_1 = 0$$

$$\text{B.C.2: } 0 = - \frac{\mu v_{z,\max}^2 R^4}{4k R^2} - 0 + C_2 \rightarrow C_2 = T_0 + \frac{\mu v_{z,\max}^2}{4k}$$

$$T(r) - T_0 = \frac{\mu v_{z,\max}^2}{4k} \left[ 1 - \left( \frac{r}{R} \right)^4 \right]$$

$$c) \rho \hat{c}_p v_{z, \max} \left[ 1 - \left( \frac{r}{R} \right)^2 \right] \frac{\partial T}{\partial z} = k \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + 4\mu \frac{v_{z, \max}^2}{R^2} \left( \frac{r}{R} \right)^2$$

$$\frac{\partial T}{\partial r} = 0$$

$$\rho \hat{c}_p v_{z, \max} \left[ 1 - \left( \frac{r}{R} \right)^2 \right] \frac{\partial T}{\partial z} = 4\mu \frac{v_{z, \max}^2}{R^2} \left( \frac{r}{R} \right)^2, \quad \xi = \frac{r}{R}$$

$$\rho \hat{c}_p v_{z, \max} \int_0^1 \left( 1 - \xi^2 \right) \xi d\xi \frac{dT}{dz} = \frac{4\mu v_{z, \max}^2}{R^2} \int_0^1 \xi^2 \xi d\xi$$

$$\frac{1}{4} \rho \hat{c}_p v_{z, \max} \frac{dT}{dz} = \frac{\mu v_{z, \max}^2}{R^2}$$

$$\frac{dT}{dz} = \frac{4\mu v_{z, \max}}{\rho \hat{c}_p R^2}$$

$$\int_{T_1}^T dT = \frac{4\mu v_{z, \max}}{\rho \hat{c}_p R^2} \int_0^z dz$$

$$T - T_1 = \frac{4\mu v_{z, \max}}{\rho \hat{c}_p R^2} z + f(r), \quad \frac{dT}{dz} = \frac{4\mu v_{z, \max}}{\rho \hat{c}_p R^2}, \quad \frac{\partial T}{\partial r} = \frac{\partial f}{\partial r}, \quad \xi = \frac{r}{R}$$

$$\frac{\partial}{\partial \xi} \left( \xi \frac{\partial f}{\partial \xi} \right) = \frac{4\mu v_{z, \max}^2}{k} (1 - 2\xi^2) \xi$$

integrate:-

$$\xi \frac{\partial f}{\partial \xi} = \frac{4\mu v_{z, \max}^2}{k} \int (1 - 2\xi^2) \xi d\xi + C_1$$

integrate:

$$f = \frac{2\mu v_{z, \max}^2}{k} \int (\xi - \xi^3) d\xi = \frac{\mu v_{z, \max}^2}{k} \left( \xi^2 - \frac{1}{2} \xi^4 \right) + C_2$$

After Applying B.C's:

$$T - T_1 = \frac{4\mu v_{z, \max}}{\rho \hat{c}_p R^2} z + \frac{\mu v_{z, \max}^2}{k} \left[ \left( \frac{r}{R} \right)^2 - \frac{1}{2} \left( \frac{r}{R} \right)^4 - \frac{1}{4} \right]$$

**11B.4. Heat conduction in a spherical shell** (Fig. 11B.4). A spherical shell has inner and outer radii  $R_1$  and  $R_2$ . A hole is made in the shell at the north pole by cutting out the conical segment in the region  $0 \leq \theta \leq \theta_1$ . A similar hole is made at the south pole by removing the portion  $(\pi - \theta_1) \leq \theta \leq \pi$ . The surface  $\theta = \theta_1$  is kept at temperature  $T = T_1$ , and the surface at  $\theta = \pi - \theta_1$  is held at  $T = T_2$ . Find the steady-state temperature distribution, using the heat conduction equation.

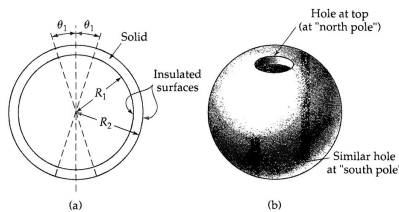


Fig. 11B.4. Heat conduction in a spherical shell: (a) cross section containing the z-axis; (b) view of the sphere from above.

Spherical coordinates:

$$T = T(\theta)$$

#### §B.9 THE EQUATION OF ENERGY FOR PURE NEWTONIAN FLUIDS WITH CONSTANT<sup>a</sup> $\rho$ AND $k$

$$[\rho \hat{C}_p DT/Dt = k \nabla^2 T + \mu \Phi_v]$$

Cartesian coordinates ( $x, y, z$ ):

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v \quad (\text{B.9-1})^b$$

Cylindrical coordinates ( $r, \theta, z$ ):

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + v_z \frac{\partial T}{\partial z} \right) = k \left[ \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \Phi_v \quad (\text{B.9-2})^b$$

Spherical coordinates ( $r, \theta, \phi$ ):

$$\rho \hat{C}_p \left( \frac{\partial T}{\partial t} + v_r \frac{\partial T}{\partial r} + \frac{v_\theta}{r} \frac{\partial T}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial T}{\partial \phi} \right) = k \left[ \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} \right] + \mu \Phi_v \quad (\text{B.9-3})^b$$

<sup>a</sup> This form of the energy equation is also valid under the less stringent assumptions  $k = \text{constant}$  and  $(\partial \ln \rho / \partial \ln T)_p Dp/Dt = 0$ . The assumption  $\rho = \text{constant}$  is given in the table heading because it is the assumption more often made.

<sup>b</sup> The function  $\Phi_v$  is given in §B.7. The term  $\mu \Phi_v$  is usually negligible, except in systems with large velocity gradients.

$$0 = \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial T}{\partial \theta} \right)$$

$$\frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial T}{\partial \theta} \right) = 0$$

$$\text{B.C.1: } \theta = \theta_1, T = T_1$$

$$\text{B.C.2: } \theta = \pi - \theta_1, T = T_2$$

integrate eqn:-

$$\sin \theta \frac{\partial T}{\partial \theta} = C_1 \rightarrow \frac{\partial T}{\partial \theta} = \frac{C_1}{\sin \theta}$$

integrate:

$$T = C_1 \ln | \tan \frac{1}{2} \theta | + C_2$$

Apply Boundary conditions:-

$$\Theta(\theta) = \frac{T - T_1}{T_2 - T_1}$$

$$T - T_1 = C_1 \ln \frac{\tan \frac{1}{2} \theta}{\tan \frac{1}{2} \theta_1} \quad \text{and} \quad T_2 - T_1 = C_1 \ln \frac{\tan \frac{1}{2} (\pi - \theta_1)}{\tan \frac{1}{2} \theta_1}$$

$$\Theta(\theta) = \frac{C_1 \ln \frac{\tan \frac{1}{2} \theta}{\tan \frac{1}{2} \theta_1}}{C_1 \ln \frac{\tan \frac{1}{2} (\pi - \theta_1)}{\tan \frac{1}{2} \theta_1}}$$

12B.4 Heat transfer from a wall to a falling film (short contact time limit)<sup>†</sup> (Fig. 12B.4). A cold liquid film flowing down a vertical solid wall, as shown in the figure, has a considerable cooling effect on the solid surface. Estimate the rate of heat transfer from the wall to the fluid for such short contact times that the fluid temperature changes appreciably only in the immediate vicinity of the wall.

(a) Show that the velocity distribution in the falling film, given in §2.2, may be written as  $v_x = v_{x,\max} [2y(\delta) - y^2/\delta^2]$ , in which  $v_{x,\max} = \rho g \delta^2 / 2\mu$ . Then show that in the vicinity of the wall the velocity is a linear function of  $y$  given by

$$v_x \approx \frac{\rho g \delta}{\mu} y \quad (12B.4-1)$$

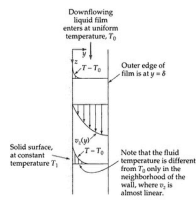


Fig. 12B.4 Heat transfer to a film falling down a vertical wall.

<sup>†</sup>R. L. Pigford, *Chemical Engineering Progress Symposium Series*, 51, No. 17, 79-92 (1955); Robert Lamar Pigford (1917-1988), who taught at both the University of Delaware and the University of California in Berkeley, researched many aspects of diffusion and mass transfer; he was the founding editor of *Industrial and Engineering Chemistry Fundamentals*.

## Chapter 12 Temperature Distributions with More Than One Independent Variable

(b) Show that the energy equation for this situation reduces to

$$\rho C_p v_x \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial y^2} \quad (12B.4-2)$$

List all the simplifying assumptions needed to get this result. Combine the preceding two equations to obtain

$$y \frac{\partial^2 T}{\partial y^2} = \beta \frac{\partial^2 T}{\partial y^2} \quad (12B.4-3)$$

in which  $\beta = \mu k / \rho C_p g \delta^3$ .

(c) Show that for short contact times we may write as boundary conditions

$$\text{B.C. 1: } T = T_0 \quad \text{for } z = 0 \quad \text{and } y > 0 \quad (12B.4-4)$$

$$\text{B.C. 2: } T = T_0 \quad \text{for } y = \infty \quad \text{and } z \text{ finite} \quad (12B.4-5)$$

$$\text{B.C. 3: } T = T_1 \quad \text{for } y = 0 \quad \text{and } z > 0 \quad (12B.4-6)$$

Note that the true boundary condition at  $y = \delta$  is replaced by a fictitious boundary condition at  $y = \infty$ . This is possible because the heat is penetrating just a very short distance into the fluid.

(d) Use the dimensionless variables  $\Theta(\eta) = (T - T_0)/(T_1 - T_0)$  and  $\eta = y/\sqrt{9\beta z}$  to rewrite the differential equation as (see Eq. C.1-9):

$$\frac{d^2 \Theta}{d\eta^2} + 3\eta \frac{d\Theta}{d\eta} = 0 \quad (12B.4-7)$$

Show that the boundary conditions are  $\Theta = 0$  for  $\eta = \infty$  and  $\Theta = 1$  at  $\eta = 0$ .

(e) In Eq. 12B.4-7, set  $\partial \Theta / \partial \eta = \eta$  and obtain an equation for  $\eta(\eta)$ . Solve that equation to get  $\partial \Theta / \partial \eta = \eta(\eta) = C_1 \exp(-\eta^2/2)$ . Show that a second integration and application of the boundary conditions give

$$\Theta = \frac{\int_0^\eta \exp(-\eta'^2/2) d\eta'}{\int_0^\infty \exp(-\eta'^2/2) d\eta'} = \frac{1}{\sqrt{\pi}} \int_0^\eta \exp(-\eta'^2/2) d\eta' \quad (12B.4-8)$$

(f) Show that the average heat flux to the fluid is

$$q_{\text{ave}} = \frac{3}{2} \frac{(9\beta L)^{-1/2}}{\sqrt{\pi}} k (T_1 - T_0) \quad (12B.4-9)$$

where use is made of the Leibnitz formula in §C.3.

$$a) \quad v_z = v_{z,\max} \left[ \frac{2y}{\delta} - \left( \frac{y}{\delta} \right)^2 \right]$$

$$v_{z,\max} = \frac{\rho g \delta^2}{2\mu}$$

$$v_z = \frac{\rho g \delta^2}{2\mu} \left[ \frac{2y}{\delta} - \frac{y^2}{\delta^2} \right]$$

$$v_z = \frac{\rho g}{\mu} \left( \delta y - \frac{y^2}{2} \right)$$

Near the wall  $y \ll \delta \quad y^2 \approx 0$

$$v_z = \frac{\rho g}{\mu} \delta y$$

$$b) \quad \rho C_p \left( v_z \frac{\partial T}{\partial z} + v_y \frac{\partial T}{\partial y} \right) = k \left( \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right)$$

$$\rho C_p v_z \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial y^2}$$

$$\rho C_p \frac{\rho g \delta}{\mu} y \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial y^2} \quad , \quad \beta = \frac{\mu k}{\rho^2 C_p g \delta}$$

$$y \frac{\partial T}{\partial z} = \beta \frac{\partial^2 T}{\partial y^2}$$

$$c) \quad \text{B.C. 1: } @ z=0 \quad T=T_0$$

$$\text{B.C. 2: } @ y=\infty \quad T=T_0$$

$$\text{B.C. 3: } @ y=0 \quad T=T_1$$

$$d) \quad \Theta = \frac{T - T_0}{T_1 - T_0} \quad \eta = y \sqrt{\frac{9\beta}{z}}$$

$$\frac{\partial^2 \Theta}{\partial \eta^2} + 3\eta^2 \frac{\partial \Theta}{\partial \eta} = 0$$

$$\text{B.C. 1: } \Theta(0) = 1$$

$$\text{B.C. 2: } \Theta(\infty) = 0$$

$$e) \quad p = \frac{d\Theta}{d\eta}$$

$$\frac{\partial p}{\partial \eta} + 3\eta^2 p = 0$$

$$\frac{1}{p} \frac{dp}{d\eta} = -3\eta^2 \rightarrow \ln p = -\eta^3 \rightarrow \frac{dp}{d\eta} = C_1 e^{-\eta^3} \rightarrow \Theta = C_1 \int e^{-\eta^3} d\eta + C_2$$

Apply Boundary Conditions:-

$$\Theta(\eta) = \frac{\int_0^\eta e^{-\eta'^3} d\eta'}{\int_0^\infty e^{-\eta'^3} d\eta'}$$

$$f) \quad q = -k \left( \frac{\partial T}{\partial y} \right)_{y=0}$$

$$q_{\text{ave}} = \frac{3}{2} \left( \frac{9\beta L}{T^{4/3}} \right)^{-1/3} k (T_1 - T_0)$$

A solid sphere of radius  $R$  is initially at a uniform temperature  $T_0$ . At time  $t = 0$ , the temperature of the sphere's surface is suddenly raised to  $T_1$ . Assume the density, heat capacity, and thermal conductivity of the solid are constant.

- 1- Write down the partial differential equation governing the transient temperature profile within the sphere.
- 2- Rewrite the differential equation in terms of dimensionless variables:  $\Theta = \frac{T_1 - T}{T_1 - T_0}$ ,  $\eta = \frac{r}{R}$ , and  $\tau = \frac{\alpha t}{R^2}$ .
- 3- Solve the dimensionless partial differential equation using the method of separation of variables. Express the dimensionless temperature profile as an infinite series and evaluate the integration constants.
- 4- Plot the dimensionless temperature as function of dimensionless radius for  $\tau = 0.1$ .
- 5- Modify the provided finite difference code for the spherical coordinates and reproduce the above plot.

$$1) \frac{\partial T}{\partial t} = K \left( \frac{1}{r^2} \left( \frac{\partial}{\partial r} \left( r^2 \frac{\partial T}{\partial r} \right) \right) \right)$$

$$T(r, 0) = T_0$$

$$T(R, t) = T_1$$

$$r=0 \rightarrow \frac{\partial T}{\partial r} = 0$$

$$2) \frac{\partial}{\partial r} = \frac{1}{R} \frac{\partial}{\partial \eta}$$

$$\frac{\partial \Theta}{\partial \tau} = \frac{1}{\eta^2} \frac{\partial}{\partial \eta} \left( \eta^2 \frac{\partial \Theta}{\partial \eta} \right)$$

$$\Theta(\eta, 0) = 1$$

$$\Theta(1, \tau) = 0$$

$$\frac{\partial \Theta}{\partial \eta} = 0 \text{ @ } \eta = 0$$

$$3) \Theta(\eta, \tau) = R(\eta) G(\tau)$$

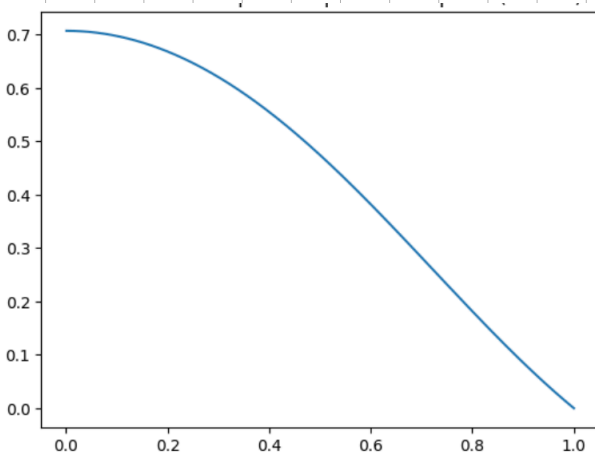
$$R \frac{dG}{d\tau} = G \frac{1}{\eta^2} \frac{d}{d\eta} \left( \eta^2 \frac{dR}{d\eta} \right)$$

$$\frac{1}{G} \frac{dG}{d\tau} = \frac{1}{R} \frac{1}{\eta^2} \frac{d}{d\eta} \left( \eta^2 \frac{dR}{d\eta} \right)$$

$$\frac{dG}{d\tau} = -\lambda^2 G$$

$$G = e^{-\lambda^2 \tau}$$

$$\Theta(\eta, \tau) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\pi} \sin(n\pi\eta) e^{-n^2\pi^2\tau}$$



```
# Spherical coordinates finite difference solution

tau = 0.1 # final dimensionless time

eta = np.linspace(0,1,100) # dimensionless radius
d_eta = eta[1]-eta[0]

theta = np.ones(eta.size) # initial condition
theta[-1] = 0 # surface BC

d_theta = np.zeros(eta.size)

d_tau = 0.00001
tau_i = 0

while tau_i < tau:
    for i in range(1,eta.size-1):
        second = (theta[i+1] - 2*theta[i] + theta[i-1]) / d_eta**2
        first = (theta[i+1] - theta[i-1]) / (2*d_eta)
        d_theta[i] = second + (2/eta[i])*first

        tau_i += d_tau
        theta = theta + d_theta*d_tau

    # center symmetry BC
    theta[0] = theta[1]

    # surface BC
    theta[-1] = 0

fig, ax = plt.subplots()
ax.plot(eta, theta, label=r"$\tau$=%.2f" % tau)

plt.legend()
plt.xlim(0,1)
plt.ylim(0,1)

ax.set_title("Finite Difference Solution (Sphere)")
ax.set_xlabel(r"$\eta$")
ax.set_ylabel(r"$\Theta$")

ax.grid()
ax.minorticks_off()
```